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Polyribonucleotides Containing a Phosphorothioate Backbone

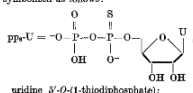
Fritz ECKSTEIN and Hans GINDL

Max-Planck-Institut für Experimentelle Medizin, Abteilung Chemie, Göttingen

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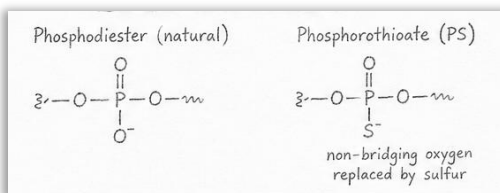
The enzymatic synthesis of two polyribonucleotides, poly r(A₃U) and poly r(A₂U) containing a sugar-phosphorothioate instead of a sugar-phosphate backbone, is described. The formation and thermal stability of double-stranded polymers is not impaired by this modification. They can also serve as templates for RNA-polymerase. Single stranded polymers can serve as messengers. Both double and single-stranded polymer exhibit some degree of resistance to nucleases.

Unusual Abbreviations. Following recommendations by Dr. W. E. Cohn (Director of the National Academy of Sciences—National Research Council, Office of Biochemical Nomenclature, Oak Ridge National Laboratory, Oak Ridge, Tennessee, U.S.A.) the phosphorothioate-containing compounds are symbolized as follows:



Recently we described [2] the synthesis and some properties of a polyribonucleotide, poly r(A₃U) in which every other phosphate was replaced by a phosphorothioate group. This modified polynucleotide was degraded by nucleases at a considerably slower rate than the unmodified polynucleotide, poly r(A₃U). The present investigation was undertaken to determine the effects of replacement of all phosphate groups in a polynucleotide by phosphorothioate. Parts of this work have been published in a preliminary report [1].

The importance of backbone chemistry in oligonucleotides dates to foundational work by Eckstein and Gindl (1970), who demonstrated that replacing a non-bridging oxygen in the phosphate backbone with sulfur (phosphorothioate) produced polyribonucleotide that retained the ability to form stable duplexes and serve as templates for RNA polymerase, while exhibiting enhanced resistance to nuclease degradation. This early work established the principle that backbone modification could preserve biological function while improving stability.



Today, backbone engineering remains a central parameter in oligonucleotide design, enabling control over stability, protein interactions, and pharmacokinetics. Phosphorothioate (PS) remains universal across ASO and many siRNA designs to improve stability, delivery, and pharmacological activity. The natural phosphodiester (PO) backbone is retained in some RNA-based modalities, including siRNA and mRNA (Egli and Manoharan, 2023; FDA, 2016; Rodrigues and Yokota, 2023).